UCLA Math 131A, Winter 2025
Practice Final

Instructor: Linfeng Li

Name: ______________________________

Student ID: _________________________

Please read all of the following rules before proceeding.

- The exam worth 70 points in total.
- Show all your work, and precisely state any theorems that you use.
- Calculators, notes, and textbooks are not allowed, but you may bring one two-sided cheat sheet.
- Good luck!

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
5	10	
6	10	
7	10	
Total:	70	

1. (10 points) Let $a > -1$ be a real number. Prove that

$$(1+a)^n \geq 1 + na$$

for all $n \in \mathbb{N}$.

2. (10 points) Let f be a continuous function on $\mathbb{R}$ such that $f(r) = r^2$ for all rational number r. Prove that $f(x) = x^2$ for all $x \in \mathbb{R}$.

3. Let $(s_n)_{n\in\mathbb{N}}$ be an increasing sequence. Suppose that there exists a subsequence $(s_{n_k})_{k\in\mathbb{N}}$ such that $(s_{n_k})_{k\in\mathbb{N}} \to s$ for some $s \in \mathbb{R}$.
   (a) (5 points) Prove that $s_n \leq s$ for all $n \in \mathbb{N}$.
   (b) (5 points) Prove that $(s_n)_{n\in\mathbb{N}}$ converges to s.

4. (10 points) Let f be a function defined on $\mathbb{N}$. Prove that f is uniformly continuous on $\mathbb{N}$.

5. (10 points) Prove that $f(x) = |x|$ is not differentiable at $x = 0$ by definition.

6. (a) (5 points) Prove that
$$\frac{\sin(x)}{x} \geq \frac{\sqrt{2}}{2}$$
for all $x \in (0, \pi/4)$.

(b) (5 points) Prove that the series
$$\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$$
diverges. *(Hint: use part (a))*

7. Let $f : [0, 1] \to \mathbb{R}$ be defined as

$$f(x) = \begin{cases} 1, & x = \frac{1}{k} \text{ for } k \in \mathbb{N} \\ 0, & \text{otherwise.} \end{cases}$$

In other words, $f(x) = 1$ for $x = 1, \frac{1}{2}, \frac{1}{3}, \ldots$ and $f(x) = 0$ otherwise.

(a) (5 points) Let the partition $P = \{0, \frac{1}{2}, 1\}$. Compute $L(f, P)$ and $U(f, P)$.

(b) (5 points) Prove that f is integrable and $\int_0^1 f(x)\,dx = 0$. *(Hint: Find a partition P of $[0, 1]$ such that $U(f, P) - L(f, P) < \epsilon$)*